# Evaluating LLM Performance for Cyberattack Anomaly Detection in O-RAN: A Zero-shot and Few-shot Analysis*

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**Abstract**

This study systematically evaluates the security event analysis capabilities of Large Language Models (LLMs) within the O-RAN environment. Four commercial LLMs—GPT-4o, GPT-4o-mini, GPT-4.1-mini, and Claude Sonnet 4.6—were tested on O-CU network layer and O-DU radio layer data using zero-shot and few-shot strategies. The experimental results yielded three key findings. First, the study demonstrated the semantic limitations of each data layer and the dramatic impact of few-shot prompting. In the O-CU layer, where prior knowledge is applicable, meaningful performance (F1 0.51–0.86) was achieved even with zero-shot learning. However, the models failed to detect threats in the O-DU layer (F1 ≈ 0.0), which consists of domain-specific data. Notably, providing only three examples caused the O-DU F1 score to surge up to 0.95, suggesting significant potential for performance improvement through future fine-tuning. Second, the models confirmed capabilities surpassing existing deep learning in specific attack detection. In O-CU web attack detection (GPT-4.1-mini, F1 0.99) and Probe detection (Claude Sonnet 4.6, F1 0.82), the LLMs significantly outperformed the accuracy of traditional CNNs. However, they exhibited common limitations in the detailed classification of attacks with similar traffic structures, such as DoS and DDoS. Third, the study proposes an optimal operating environment based on interpretive capabilities that complement deep learning. Although the LLMs exceeded real-time processing constraints (1,000ms), they demonstrated a unique ability to generate specific attack reasoning and response recommendations beyond simple classification. This research suggests that LLMs are suitable as post-analysis tools to complement deep learning-based detection within the Non-RT RIC environment.

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# 1 Introduction

The open interfaces and cloud-based design of O-RAN have enhanced the flexibility of Radio Access Networks (RAN), but they have simultaneously significantly expanded the attack surface. According to O-RAN Alliance WG11, approximately 60% of identified threats are related to Denial-of-Service (DoS) attacks and performance degradation. To counter these threats, previous studies have demonstrated that deep learning-based IDS xApps deployed in Near-RT RIC can achieve high detection accuracy [1]. However, deep learning models require domain-specific training data and fail to provide explanations for detection results, making it difficult for operators to make immediate situational judgments.

LLMs present a fundamentally different approach to security analysis. They can perform classification using only natural language prompts without separate training, while simultaneously generating human-readable explanations and response recommendations. Although Moore et al. achieved 99% accuracy in O-RAN detection using a fine-tuned Gemma 2 model, there is a lack of research systematically evaluating general-purpose LLMs on dual-layer data from O-CU and O-DU using various prompting strategies [2].

This paper evaluates four LLMs—OpenAI's GPT-4o, GPT-4o-mini, and GPT-4.1-mini, along with Anthropic's Claude Sonnet 4.6—under zero-shot and few-shot settings, providing the following contributions: First, it identifies gaps across data layers and demonstrates the effectiveness of real-time knowledge injection through few-shot prompting. Second, it identifies performance advantages over existing deep learning in detecting specific threats. Third, by analyzing the unique interpretive capabilities of LLMs that deep learning cannot provide, it presents the practical value of generating attack type reasoning and response recommendations in an operational environment.

# 2 Experimental Results and Analysis

This study utilized the NetsLab 5G O-RAN IDD dataset to evaluate the performance of LLM-based security analysis in an O-RAN environment. In particular, the deep learning-based IDS xApp proposed in a previous study (Yu et al., 2025) was used as a baseline for performance comparison to analyze the detection capabilities of LLMs from various perspectives [1,3].

**Binary Detection Performance by Data Layer:** The detection behavior of LLMs differed fundamentally between the network layer and the radio layer data. In the O-CU network layer, zero-shot LLMs showed detection performance ranging from F1 0.51 to 0.86 by utilizing network knowledge acquired during pre-training, such as port numbers, protocols, and packet sizes. When few-shot prompting was applied, the performance of all four models rose to F1 0.92–0.96, approaching the highest deep learning performance of 0.99. A completely different pattern emerged in the O-DU radio layer. In a zero-shot environment, all GPT models classified all 480 samples as normal, resulting in an F1 score of 0.0. This indicates that LLMs have no prior understanding of radio telemetry values such as RSRP, SINR, and MCS. However, upon applying few-shot prompting with only three examples per category, Claude Sonnet 4.6 improved by +0.77 (from F1 0.18 to 0.95), and GPT-4o improved by +0.93 (from 0.0 to 0.93). This demonstrates that a small number of examples can effectively inject domain knowledge—linking specific numerical patterns to certain attacks—into the LLM in real-time. Comparing the degree of improvement, the average ΔF1 of the radio layer was +0.85, which is more than three times larger than the +0.28 of the network layer. This suggests that the effect of few-shot prompting is maximized in domains where the LLM lacks prior knowledge.

**Detailed Detection Characteristics by Multi-class and Attack Type:** Macro F1 measurements for six traffic types (Normal, DoS, DDoS, Probe, Brute Force, and Web Attack) showed that with few-shot prompting, performance improved to 0.50–0.69 for O-CU and up to 0.85 for O-DU (based on

Claude Sonnet). LLMs showed strengths in attack types with clear signatures. In web attack detection, GPT-4.1-mini recorded a few-shot F1 of 0.99, significantly exceeding the CNN's 0.67, and in Probe detection, Claude Sonnet 4.6 achieved an F1 of 0.82, higher than the CNN's 0.45. Conversely, distinguishing between DoS and DDoS was a common weakness for all LLMs. Because both attacks have similar structures generating large volumes of traffic, it is difficult to capture the difference between single-source and distributed-source attacks using only numerical features.

**LLM Interpretive Capability and Speed-Accuracy Trade-off:** While all LLMs exceeded the 1,000ms latency constraint of Near-RT RIC, they generated unique value that deep learning cannot provide during this latency period. When classifying traffic as an attack, the LLMs did not merely provide a binary judgment but output technical explanations referencing specific feature patterns along with concrete response measures. For example, they provided structured analysis such as: "An extremely short duration and dst_bytes of 0 indicate a port scan pattern; it is recommended to block the source IP and monitor for further scanning activity." This directly addresses the issue where deep learning-based IDS xApps provide only binary alerts, making it difficult for operators to understand the context. Even the fastest model, GPT-4.1-mini, incurred 1.6 times the latency, and in few-shot environments, latency increased by 1.4 to 2.5 times due to the increase in example tokens. A clear trade-off was observed between Claude Sonnet 4.6, which had the highest accuracy, and GPT-4.1-mini, which had an advantage in speed. Consequently, few-shot-based LLM analysis is more suitable as a post-analysis tool for in-depth analysis and explanation generation within a Non-RT RIC environment rather than as a real-time defense system.

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